

1. The following data is related to the measurement of cholesterol content (%) performed in 4 different laboratories w.r.t 4 diet foods. The design adopted is a **Randomized Block design**.

Analyze the data and comment on your result

[Given $F_{0.05} = 3.86$ for (3,6) D.F]

	Laboratory			
	I	II	III	IV
A	4	5	6	4
B	3	4	5	8
C	5	3	7	5
D	2	3	4	2

ANS:

No. of Rows, R = 4

No. of Columns, C = 4

If R and C are Equal to 'n',

Order of the square, n = 4

No. of observation, N (Rows $\times$ Columns) = n^2 or ($R \times C$)
= 16

I	II	III	IV	Σy_{ij}
4	5	6	4	
3	4	5	8	
5	3	7	5	
2	3	4	2	70
14	15	22	19	

Grand Total, $T = \Sigma y_{ij}$ = 70

Correction Factor, C. F = $(\Sigma y_{ij})^2 / N$ = $490/16$
= 306.25

I^2	II^2	III^2	IV^2	Σy_{ij}^2
16	25	36	16	
9	16	25	64	
25	9	49	25	
4	9	16	4	348
54	59	126	109	

Total Sum of Square, TSS

$$\begin{aligned}
 &= \sum y_{ij}^2 - C \cdot F \\
 &= 348 - 306.25 \\
 &= 41.75
 \end{aligned}$$

	I	II	III	IV	R_i	R_i^2	
A	4	5	6	4	19	361	
B	3	4	5	8	20	400	
C	5	3	7	5	20	400	
D	2	3	4	2	11	121	
C_j	14	15	22	19		1282	$(\sum R_i^2)$
C_j^2	196	225	484	361	1266		
					$(\sum C_j^2)$		

Sum of Square between Rows, SSR

$$\begin{aligned}
 &= (\sum R_i^2) / R - C \cdot F \\
 &= 1282 / 4 - 306.25 \\
 &= 320.50 - 306.25 \\
 &= 14.25
 \end{aligned}$$

Sum of Square between Columns, SSC

$$\begin{aligned}
 &= (\sum C_j^2) / C - C \cdot F \\
 &= 1266 / 4 - 306.25 \\
 &= 316.50 - 306.25 \\
 &= 10.25
 \end{aligned}$$

Sum of Square Error, SSE

$$\begin{aligned}
 &= TSS - SSR - SSC \\
 &= 41.75 - 14.25 - 10.25 \\
 &= 17.25
 \end{aligned}$$

Analysis:

Source of Variation	Degree of Freedom	Sum of Square	Mean Sum of Square	Observed Value	Table Value
	DF	SS	MSS = SS/DF	Fo = MSS/MSE	Fe = F(,E)
Between Rows	$R-1 = 3$	14.25	$MSR = SSR / (R-1) = 4.75$	$4.75 / 1.92 = 2.48$	$F(3,6) = 3.86$
Between Columns	$C-1 = 3$	10.25	$MSC = SSC / (C-1) = 3.42$	$3.42 / 1.92 = 1.78$	$F(3,6) = 3.86$
Error	$(n-1)(n-1) = 3 \times 3 = 9$	17.25	$MSE = SSE / \text{Error} = 1.92$		
Total	$N-1 = n^2 - 1 = 15$				

Inference:

<i>Rows Diet Food</i>	<i>Columns laboratory</i>
$F_0 = 2.47$ $F_e = 3.86$ Since $F_0 < F_e$, we infer that there is no significant difference in cholestrol % between Diet Foods.	$F_0 = 1.78$ $F_e = 3.86$ Since $F_0 < F_e$, we infer that there is no significant difference in cholestrol(%) between Laboratories

2. A company manufacturing a consumer product appointed four salesmen and observed their sales in 3 different states. The no.of products sold is given below.

<i>State</i>	<i>Salesmen</i>			
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>I</i>	22	27	38	45
<i>II</i>	28	32	40	35
<i>III</i>	25	40	36	22

The design adopted is a randomized block design. Analyse the data and comment on your result. $F_{0.05}(3,6) = 4.76$, $F_{0.05}(2,6) = 5.14$.

Solution:

Effect of salesmen and state - RBD

State Rows, R = 3

Salemen Columns, C = 4

No.of Rows, R = 3

No.of Columns, C = 4

$$\begin{aligned}\text{No. of observation, } N (\text{Rows} \times \text{Columns}) &= n^2 \text{ or } (R \times C) \\ &= 3 \times 4 = 12\end{aligned}$$

State	Salem				
	A	B	C	D	
I	22	27	38	45	Σy_{ij}
II	28	32	40	35	
III	25	40	36	22	
					390

$$\begin{aligned}\text{Grand Total, } T = \Sigma y_{ij} &= 390 \\ \text{Correction Factor, } C.F. = (\Sigma y_{ij})^2 / N &= 390^2 / 12 \\ &= 12675\end{aligned}$$

A^2	B^2	C^2	D^2	Σy_{ij}^2
484	729	1444	2025	
784	1024	1600	1225	
625	1600	1296	484	
1893	3353	4340	3734	13320

$$\begin{aligned}\text{Total Sum of Square, TSS} &= \Sigma y_{ij}^2 - C.F. \\ &= 13320 - 12675 \\ &= 645\end{aligned}$$

	A	B	C	D	R_i	R_i^2	
I	22	27	38	45	132	17424	
II	28	32	40	35	135	18225	
III	25	40	36	22	123	15129	
C_j	75	99	114	102		50778	(ΣR_i^2)
C_j^2	5625	9801	12996	10404	38826		
					(ΣC_j^2)		

$$\begin{aligned}\text{Sum of Square between Rows, SSR} &= (\Sigma R_i^2) / C - C.F. \\ &= 50778 / 4 - 12675 \\ &= 12694.5 - 12675 \\ &= 19.5\end{aligned}$$

$$\begin{aligned}\text{Sum of Square between Columns, SSC} &= (\Sigma C_j^2) / R - C.F. \\ &= 38826 / 3 - 12675\end{aligned}$$

$$= 12942 - 12675$$

$$= 267$$

$$\begin{aligned} \text{Sum of Square Error, SSE} &= TSS - SSR - SSC \\ &= 645 - 19.5 - 267 = 358.5 \end{aligned}$$

Analysis:

Source of Variation	Degree of Freedom	Sum of Square	Mean Sum of Square	Observed Value	Table Value
	DF	SS	MSS = SS/DF	Fo = MSS/MSE	Fe = F(,E)
Between Rows	R-1 = 2	19.5	MSR=SSR/(R-1) = 9.75	4.75/1.92 = 0.1632	F(3,6)= 5.14
Between Columns	C-1 = 3	267	MSC=SSC/(R-1) = 89	3.42/1.92 = 1.4895	F(3,6)= 4.76
Error	Total - DFR-DFC= 6	358.5	MSE=SSE/Error = 59.75		
Total	N-1 = RC-1 = 11				

Inference:

Rows Diet Food	Columns laboratory
$F_0 = 0.1632$ $F_e = 5.14$ Since $F_0 < F_e$, we infer that there is no significant difference in No.of products manufactured between States.	$F_0 = 1.4895$ $F_e = 4.76$ Since $F_0 < F_e$, we infer that there is no significant difference in No.of products manufactured between States.

3. An oil company is interested in testing four different blends of gasoline for the fuel efficiency by controlling the variability of four different drivers and four models of cars. The fuel efficiency was measured as kilometer per litre. After driving the cars over a standard clause. Data is presented in 4×4 Latin square Design.

Solution:

No. of Rows, R $= 4$

No. of Columns, C $= 4$

No. of observation, N (Rows $\times$ Columns) $= R \times C$

A(13)	D(9)	C(10)	B(12)
B(12)	C(10)	A(13)	D(8)
C(10)	B(12)	D(9)	A(13)
D(9)	A(14)	B(12)	C(10.5)

$$= 4 \times 4 = 16$$

A(13)	D(9)	C(10)	B(12)	Σy_{ij}
B(12)	C(10)	A(13)	D(8)	
C(10)	B(12)	D(9)	A(13)	
D(9)	A(14)	B(12)	C(10.5)	
				176.5

Grand Total, $T = \Sigma y_{ij}$ $= 176.5$

Correction Factor, C. F = $(\Sigma y_{ij})^2 / N$ $= 175.5^2 / 16$

$$= 1947.02$$

A(169)	D(81)	C(100)	B(144)	Σy_{ij}^2
B(144)	C(100)	A(169)	D(64)	
C(100)	B(144)	D(81)	A(169)	
D(81)	A(196)	B(144)	C(110.25)	
494	521	494	487.25	1996.25

Total Sum of Square, TSS

$$\begin{aligned}
 &= \sum y_{ij}^2 - C.F \\
 &= 1996.25 - 1947.02 \\
 &= 49.23
 \end{aligned}$$

					R_i	R_i^2	
	A(13)	D(9)	C(10)	B(12)	44	1936	
	B(12)	C(10)	A(13)	D(8)	43	1849	
	C(10)	B(12)	D(9)	A(13)	44	1936	
	D(9)	A(14)	B(12)	C(10.5)	45.5	2070.25	
C_j	44	45	44	43.5		7791.25	$(\sum R_i^2)$
C_j^2	1936	2025	1936	1892.25	7789.25		
					$(\sum C_j^2)$		

Sum of Square between Rows, SSR

$$\begin{aligned}
 &= (\sum R_i^2) / C - C.F \\
 &= 7791.25 / 4 - 1947.02 \\
 &= 1947.81 - 1947.02 \\
 &= 0.79
 \end{aligned}$$

Sum of Square b/n Columns, SSC

$$\begin{aligned}
 &= (\sum C_j^2) / R - C.F \\
 &= 7789.25 / 4 - 1947.02 \\
 &= 1947.31 - 1947.02 \\
 &= 0.29
 \end{aligned}$$

Sum of Square b/n Treatment

$$= (\sum T_i^2) / n - C.F$$

$$\begin{aligned}
 \text{Where, } (\sum T_i^2) &= (\sum A)^2 + (\sum B)^2 + (\sum C)^2 + (\sum D)^2 \\
 &= (53)^2 + (48)^2 + (40.5)^2 + (35)^2 \\
 &= 7978.25
 \end{aligned}$$

$$\begin{aligned}
 &= 7978.25 / 4 - 1947.02 \\
 &= 1994.56 - 1947.02 \\
 &= 47.54
 \end{aligned}$$

Sum of Square Error, SSE

$$\begin{aligned}
 &= TSS - SSR - SSC - SST \\
 &= 49.23 - 0.79 - 0.29 - 47.54 \\
 &= 0.61
 \end{aligned}$$

Analysis:

Source of Variation	Degree of Freedom	Sum of Square	Mean Sum of Square	Observed Value	Table Value
	DF	SS	MSS = SS/DF	Fo = MSS/MSE	Fe = F(,E)
Between Rows	R-1 = 3	0.79	MSR=SSR/(R-1) = 0.2633	0.2633/0.1017 = 2.59	F(3,6)= 4.76
Between Columns	C-1 = 3	0.29	MSC=SSC/(R-1) = 0.0967	0.0967/0.1017 = 0.95	F(3,6)= 4.76
Between Treatment	T-1 = 3	47.54	MST=SST/(T-1) = 15.8467	15.8467/0.1017 = 155.82	F(3,6)= 4.76
Error	N - DFR - DFC - DFT = 6	0.61	MSE=SSE/Error = 0.1017		
Total	N-1 = RC-1 = 15				

Inference:

Rows	Columns	Treatment
$F_0 = 2.59$ $F_e = 4.76$ Since $F_0 < F_e$, we infer that there is no significant difference between Rows.	$F_0 = 0.95$ $F_e = 4.76$ Since $F_0 < F_e$, we infer that there is no significant difference between Columns.	$F_0 = 155.82$ $F_e = 4.76$ Since $F_0 > F_e$, we infer that there is significant difference between Treatments.

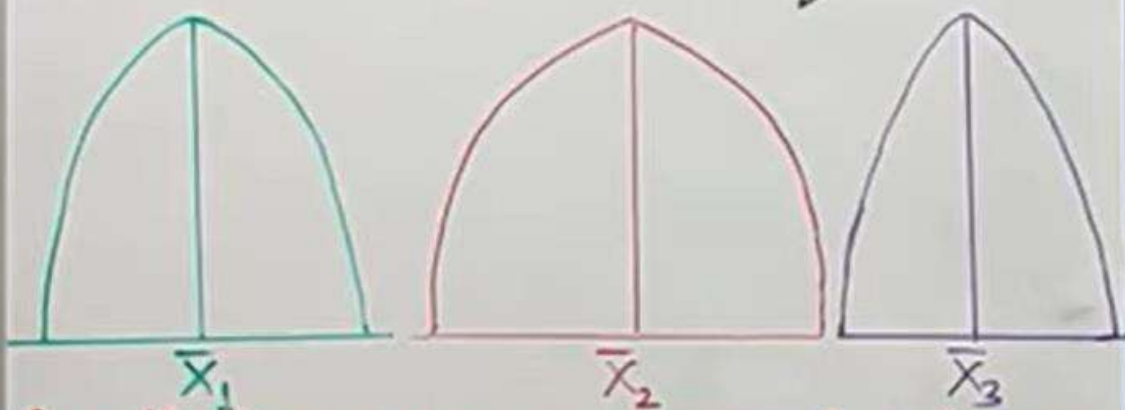
Analysis of Variance

Introduction &

- ANOVA is developed by R. A. Fisher in 1920

Variance:- It is defined as the expectation of the squared deviation of a random variable from its mean. i.e. σ^2 or s^2

- for comparison of more than two population or population having more than two subgroups we will use ANOVA technique.



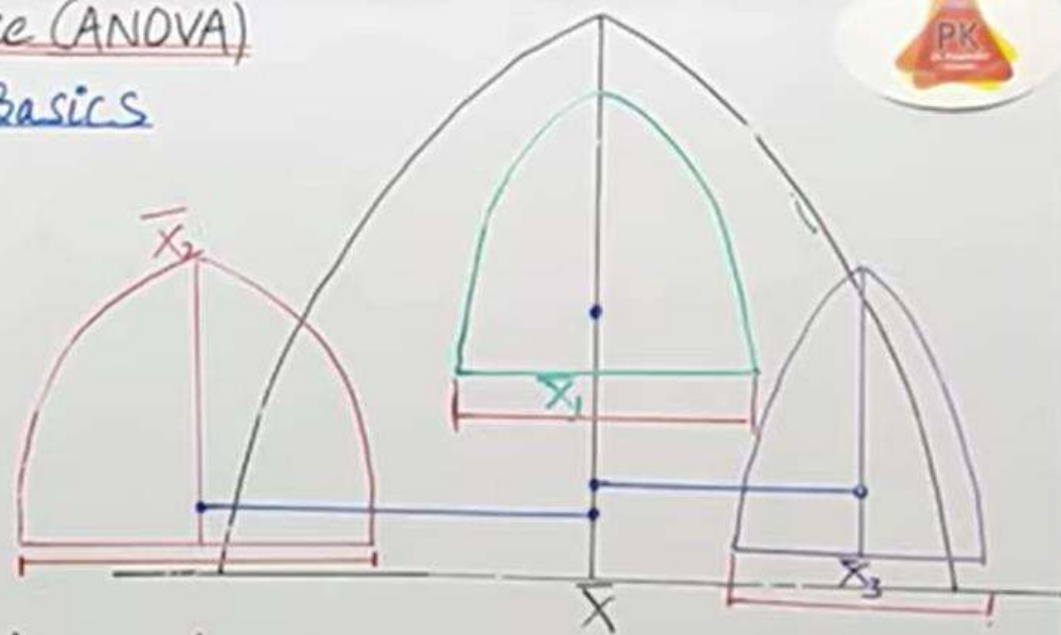
Do all these 3 means are coming from the same population?

$$\text{ANOVA} = \frac{\text{Variability between the Means}}{\text{Variability within the distribution}}$$

$$\text{Total variance} = \text{variability between the means} + \text{variability within the distribution.}$$

ANOVA (ANOVA)

Basics



Assumptions:-

- ① Each population is having normal distribution.
- ② The population from which the sample are drawn have the equal variance.
i.e. $S_1^2 = S_2^2 = S_3^2 = \dots = S_k^2$ for k samples.
- ③ Each sample is drawn randomly & they are independent.

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_n$$

$$H_a: \mu_1 \neq \mu_2 \neq \mu_3 = \dots \neq \mu_n$$

Classification $\left\{ \begin{array}{l} \text{One factor - One way ANOVA} \\ \text{Two factor - Two way ANOVA} \end{array} \right.$